

Notes on Spectral (Fourier) Method of Integration

PHYS4840 / Joyce / Classs 26

Overview

The **spectral method** is a numerical technique for solving differential equations by expanding functions in terms of global basis functions. When using a **Fourier basis**, these basis functions are sines and cosines (or complex exponentials), and the function is expressed as a Fourier series:

$$f(x) \approx \sum_{k=-N}^N \hat{f}_k e^{ikx}$$

Here, $\hat{f}_k$ are the Fourier coefficients, and e^{ikx} are the basis functions.

Integration Using the Fourier Spectral Method

Integration in the spectral method becomes algebraic. If a function is represented by its Fourier series:

$$f(x) = \sum_{k=-N}^N \hat{f}_k e^{ikx}$$

then its indefinite integral is:

$$F(x) = \sum_{k \neq 0} \frac{\hat{f}_k}{ik} e^{ikx} + C$$

The term for $k = 0$ corresponds to a constant in $f(x)$, and its integral becomes a linear function, which must be handled separately. The constant C is the integration constant.

Example: Integrating $f(x) = \cos(x)$

We consider the function

$$f(x) = \cos(x)$$

on the domain $[-\pi, \pi]$.

Step 1: Fourier Series

The cosine function can be written using Euler's formula:

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

So the nonzero Fourier coefficients are:

$$\hat{f}_1 = \hat{f}_{-1} = \frac{1}{2}$$

Step 2: Integrate Term-by-Term

We integrate each mode separately:

$$F(x) = \sum_{k \neq 0} \frac{\hat{f}_k}{ik} e^{ikx} = \frac{1}{2i} \left(\frac{e^{ix}}{1} - \frac{e^{-ix}}{1} \right) = \frac{1}{2i} (e^{ix} - e^{-ix}) = \sin(x)$$

So the result of integrating $\cos(x)$ is:

$$\int \cos(x) dx = \sin(x) + C$$

which matches the standard result.

Practical Implementation with FFTs

In practical numerical applications:

1. Discretize $f(x)$ on a uniform grid.
2. Compute the FFT to obtain $\hat{f}_k$.
3. Integrate in Fourier space by dividing by ik (excluding $k = 0$).
4. Apply the inverse FFT to get $F(x)$.

This approach is highly accurate for smooth periodic functions and forms the basis of spectral numerical methods used in computational physics and fluid dynamics.

FFT Frequency Array in Spectral Integration

In the Python line:

```
k = np.fft.fftfreq(N, d=L/N) * 2 * np.pi
```

we are constructing the array of **angular frequencies** (Fourier wave numbers) k , appropriate for use in Fourier spectral methods.

Breaking it Down

- N is the number of grid points (samples).
- L is the total length of the domain.
- $d = \frac{L}{N}$ is the spacing between points.
- `np.fft.fftfreq(N, d)` returns the frequency bins (in cycles per unit length):

$$\left[0, \frac{1}{L}, \frac{2}{L}, \dots, \frac{N}{2L}, -\frac{N}{2L}, \dots, -\frac{1}{L} \right]$$

for even N .

- Multiplying by 2π converts from cycles per unit length to **angular frequencies** (radians per unit length):

$$k = 2\pi\nu$$

Example: Domain $[-\pi, \pi]$ with $N = 64$

- The domain length is $L = 2\pi$.
- The grid spacing is $d = \frac{L}{N} = \frac{2\pi}{64}$.
- The frequency array becomes:

$$k = 2\pi \cdot \text{fftfreq}(64, d = 2\pi/64) \Rightarrow k = [0, 1, 2, \dots, 31, -32, \dots, -1]$$

in units of radians per unit length.

This k array is then used in spectral formulas such as:

$$f(x) = \sum_k \hat{f}_k e^{ikx}, \quad \int f(x) dx \Rightarrow \sum_{k \neq 0} \frac{\hat{f}_k}{ik} e^{ikx}$$

Why Are Some Fourier Wave Numbers Negative?

The appearance of both positive and negative wave numbers in a Fourier transform arises from the choice of basis functions. In the complex Fourier basis, we represent a function $f(x)$ using complex exponentials:

$$f(x) = \sum_{k=-N/2}^{N/2} \hat{f}_k e^{ikx}$$

Here, $k \in \mathbb{Z}$ includes both positive and negative integers, and the basis functions e^{ikx} span the full space of periodic functions over a domain.

Physical Meaning of Sign

- **Positive k :** Corresponds to waves rotating counterclockwise in the complex plane, or propagating rightward in space.
- **Negative k :** Corresponds to waves rotating clockwise, or propagating leftward.

The presence of both directions is essential for fully capturing the behavior of general functions, especially those with odd or asymmetric features.

Real-Valued Functions and Hermitian Symmetry

If $f(x)$ is real-valued, its Fourier coefficients satisfy the Hermitian symmetry:

$$\hat{f}_{-k} = \overline{\hat{f}_k}$$

This ensures that the imaginary parts cancel in the inverse transform, yielding a real function.

FFT Indexing Convention

In the discrete Fourier transform (DFT), we compute frequency components using index values $k = 0, 1, \dots, N-1$. However, because the Fourier basis functions $e^{-2\pi i k n / N}$ are periodic in k , we interpret the second half of the index range as corresponding to negative frequencies.

Specifically, the indices $k = N/2 + 1, \dots, N-1$ are reinterpreted as frequencies $k = -N+1, \dots, -1$. This allows both forward and backward wave contributions to be represented in a single array of length N . The function `np.fft.fftfreq(N)` handles this mapping automatically.

Example: FFT Index Mapping for $N = 8$

For $N = 8$, the raw DFT/FFT output corresponds to indices:

$$k = [0, 1, 2, 3, 4, 5, 6, 7]$$

However, in frequency space, these correspond to:

$$[0, 1, 2, 3, -4, -3, -2, -1] / N$$

Array Index	Interpreted Frequency k/N	Meaning
0	0	DC (mean value)
1	1/8	Positive frequency
2	2/8	Positive frequency
3	3/8	Positive frequency
4	-4/8	Nyquist frequency (real-valued)
5	-3/8	Negative frequency
6	-2/8	Negative frequency
7	-1/8	Negative frequency

This negative-index interpretation ensures that both directions of oscillation are represented in the frequency decomposition, preserving the full structure of the original function.

Summary Table

k	Physical Meaning	Basis Term
> 0	Forward wave (rightward)	e^{ikx}
$= 0$	DC component (mean)	Constant
< 0	Backward wave (leftward)	e^{ikx} for $k < 0$